5a(i)	(by Fleming's left hand) magnetic force is right So electric force is left and so left plate is lower potential	**M1** **A1**
a(ii)	$Bqv = qE \Rightarrow v = E/B = 1.50 \times 10^{4} / 0.020$ $= 7.5 \times 10^{5}\ \mathrm{m\ s^{-1}}$	**C1** **A1**
a(iii)	The ions of other speeds are deflected. Faster ions experience larger magnetic force and are deflected to the right, slower ions are deflected to the left. Only ions of certain "chosen" velocity experience equal magnetic and electric force, then remain undeflected and emerge from the slit.	B1 B1
b(i)	$Bqv = mv^{2}/r \Rightarrow r = mv/Bq = (35)(1.66 \times 10^{-27})(7.5 \times 10^{5}) / [(0.020)(1.60 \times 10^{-19})]$ $= 13.6\ \mathrm{m}$	**C1** **A1**
b(ii)	1. no effect. No effect. 2. no effect, radius is halved	**B1** **B1**
6 (a)	Root-mean-square value of an alternating current is defined as the value of an	